Thermodynamic Constitutive Theory

For the calculation of thermo-mechanical boundary value problems, we have the equations

balance of mass	1	$\dot{\varrho} = 0$
balance of linear momentum	3	$\operatorname{div} \underset{\sim}{\sigma} = \varrho(\ddot{\underset{\sim}{u}} - \underset{\sim}{b})$
balance of energy	1	$\dot{\psi} + \dot{\vartheta}\eta + \vartheta\dot{\eta} = \underset{\sim}{\sigma} \cdot \dot{\underset{\sim}{\varepsilon}} + r - \operatorname{div} \underset{\sim}{q}$
Σ	5	scalar-valued equations

Unknowns

Displacements	$\underset{\sim}{u}$	3	(Kinematics $\underset{\sim}{\varepsilon} = \operatorname{grad}_{sym} \underset{\sim}{u}$ cons.)
Stresses	$\underset{\sim}{\sigma}$	6	($\underset{\sim}{\sigma} = \underset{\sim}{\sigma}^T$ considered)
Density	ϱ	1	
Free energy	ψ	1	
Specific entropy	η	1	
Temperature	ϑ	1	
Heat flow	$\underset{\sim}{q}$	3	
Σ		16	

$\rightsquigarrow$ $16 - 5 = 11$ const. equations required

$\mapsto$ da $\dot{\varrho} = 0$

$$\psi := \psi\left(\underset{\sim}{X}, \vartheta, \dot{\vartheta}, \operatorname{grad}\vartheta, \operatorname{grad}\dot{\vartheta}, \underset{\sim}{\varepsilon}, \dot{\underset{\sim}{\varepsilon}}, \cancel{\varrho}, \cancel{\dot{\varrho}}\right)$$

$$\eta := \eta\left(\quad " \quad \right)$$

$$\underset{\sim}{\sigma} := \underset{\sim}{\sigma}\left(\quad " \quad \right)$$

$$\underset{\sim}{q} := \underset{\sim}{q}\left(\quad " \quad \right)$$

when neglecting rate-dependencies:

$$\psi := \psi(\vartheta, \operatorname{grad}\vartheta, \underset{\sim}{\varepsilon})$$
$$\eta := \eta(\quad " \quad)$$
$$\underset{\sim}{\sigma} := \underset{\sim}{\sigma}(\quad " \quad)$$
$$\underset{\sim}{q} := \underset{\sim}{q}(\quad " \quad)$$

Consider Clausius-Duhem-inequality

$$\underset{\sim}{\sigma}\cdot\dot{\underset{\sim}{\varepsilon}} - \dot{\psi} - \dot{\vartheta}s\eta - \frac{1}{\vartheta}\underset{\sim}{q}\cdot\operatorname{grad}\vartheta \geq 0$$

$$\dot{\psi} = \frac{\partial\psi}{\partial\vartheta}\dot{\vartheta} + \frac{\partial\psi}{\partial\operatorname{grad}\vartheta}\cdot\operatorname{grad}\dot{\vartheta} + \frac{\partial\psi}{\partial\underset{\sim}{\varepsilon}}\cdot\dot{\underset{\sim}{\varepsilon}}$$

$$\rightsquigarrow \underset{\sim}{\sigma}\cdot\dot{\underset{\sim}{\varepsilon}} - \frac{\partial\psi}{\partial\vartheta}\dot{\vartheta} - \frac{\partial\psi}{\partial\operatorname{grad}\vartheta}\cdot\operatorname{grad}\dot{\vartheta} - \frac{\partial\psi}{\partial\underset{\sim}{\varepsilon}}\cdot\dot{\underset{\sim}{\varepsilon}} - \dot{\vartheta}s\eta - \frac{1}{\vartheta}\underset{\sim}{q}\cdot\operatorname{grad}\vartheta \geq 0$$

$$\Leftrightarrow \left(\underset{\sim}{\sigma} - \frac{\partial\psi}{\partial\underset{\sim}{\varepsilon}}\right)\cdot\dot{\underset{\sim}{\varepsilon}} - \left(s\eta + \frac{\partial\psi}{\partial\vartheta}\right)\dot{\vartheta} - \frac{\partial\psi}{\partial\operatorname{grad}\vartheta}\cdot\operatorname{grad}\dot{\vartheta} - \frac{1}{\vartheta}\underset{\sim}{q}\cdot\operatorname{grad}\vartheta \geq 0$$

Standard argument of rational cont. mech.:

$\rightsquigarrow$
- $\underset{\sim}{\sigma} = \dfrac{\partial\psi}{\partial\underset{\sim}{\varepsilon}}$
- $s\eta = -\dfrac{\partial\psi}{\partial\vartheta}$
- $\dfrac{\partial\psi}{\partial\operatorname{grad}\vartheta} = \underset{\sim}{0}$ $\rightsquigarrow$ ψ is no function of $\operatorname{grad}\vartheta$!
- $-\dfrac{1}{\vartheta}\underset{\sim}{q}\cdot\operatorname{grad}\vartheta \geq 0$ (a priori fulfilled if $\underset{\sim}{q} \sim -\operatorname{grad}\vartheta$)

$\rightsquigarrow$ Standard model for heat flow

Fourrier law: $\underset{\sim}{q} = -\underline{K}\,\mathrm{grad}\,\vartheta$

2nd order tensor of heat conduction,
for isotropic heat conduction:

$$\underline{K} = K\,\underline{1}\,, \quad K > 0$$

$$\rightsquigarrow -\frac{1}{\vartheta}\,\underset{\sim}{q}\cdot\mathrm{grad}\,\vartheta = -\frac{1}{\vartheta}\,(-\underline{K})\,\mathrm{grad}\vartheta\cdot\mathrm{grad}\vartheta \geq 0 \;\checkmark$$

Linear thermo-elastic material behavior

Consider Taylor-expansion of energy

$$\psi(\vartheta,\underline{\varepsilon}) = \psi(\vartheta_0,\underline{\varepsilon}_0=\underline{0}) + \underbrace{\frac{\partial\psi(\vartheta_0,\underline{0})}{\partial\underline{\varepsilon}}}\cdot(\underline{\varepsilon}-\underline{0}) + \underbrace{\frac{\partial\psi(\vartheta_0,\underline{0})}{\partial\vartheta}}(\vartheta-\vartheta_0) +$$

Eigenstresses $\underline{\sigma}_0$ — Entropy in ref.-conf. $-\varrho\eta_0$

$$+\frac{1}{2}\left[\underline{\varepsilon}\cdot\left(\underbrace{\frac{\partial^2\psi(\vartheta_0,\underline{0})}{\partial\underline{\varepsilon}^2}}:\underline{\varepsilon}\right) + 2\underbrace{\frac{\partial^2\psi(\vartheta_0,\underline{0})}{\partial\underline{\varepsilon}\,\partial\vartheta}}\cdot\underline{\varepsilon}\,(\vartheta-\vartheta_0) +\right.$$

$\underline{\mathbb{C}}$ — thermal expansion tensor $-\underline{A}$

$$\left. + \underbrace{\frac{\partial^2\psi(\vartheta_0,\underline{0})}{\partial\vartheta^2}}(\vartheta-\vartheta_0)^2\right] + \underbrace{O(\underline{\varepsilon}^3,(\vartheta-\vartheta_0)^3)}$$

heat capacity $-c$

neglected for linear model, ok for $(\underline{\varepsilon}-\underline{\varepsilon}_0)$ and $(\vartheta-\vartheta_0)$ small!

- Energy for $\underline{\sigma}_0 = \underline{0}$ and $s\eta_0 = 0$:

$$\rightsquigarrow \quad \psi := \frac{1}{2}\,\underline{\varepsilon}\cdot\left(\underline{\mathbb{C}} : \underline{\varepsilon}\right) - \underline{A}\cdot\underline{\varepsilon}\,(\vartheta-\vartheta_0) - \frac{1}{2}k(\vartheta-\vartheta_0)^2$$

$$\frac{\partial^2\psi}{\partial\underline{\varepsilon}^2} = \underline{\mathbb{C}} \quad \checkmark$$

$$\frac{\partial^2\psi}{\partial\underline{\varepsilon}\partial\vartheta} = -\underline{A} \quad \checkmark$$

$$\frac{\partial^2\psi}{\partial\vartheta^2} = -c \quad \checkmark$$

- Stress

$$\underline{\sigma} = \frac{\partial\psi}{\partial\underline{\varepsilon}} = \underline{\mathbb{C}} : \underline{\varepsilon} - \underline{A}\,(\vartheta-\vartheta_0) \overset{!}{=} \underline{\mathbb{C}} : \overbrace{\left(\underline{\varepsilon} - \underline{\varepsilon}^{\text{therm}}\right)}^{\underline{\varepsilon}^{\text{elast}}}$$

Introduction of isotropic thermal expansion

$$\underline{\varepsilon}^{\text{therm}} := -\alpha_\vartheta\,(\vartheta-\vartheta_0)\,\underline{1}$$

α_ϑ: therm. expansion coefficient

$$\rightsquigarrow \quad \underline{A}\,(\vartheta-\vartheta_0) \overset{!}{=} \underline{\mathbb{C}} : \underline{\varepsilon}^{\text{therm}}$$

$$= \underline{\mathbb{C}} : \left(-\alpha_\vartheta\,(\vartheta-\vartheta_0)\,\underline{1}\right)$$

$$\Leftrightarrow \quad \underline{A} = -\alpha_\vartheta\,\underline{\mathbb{C}} : \underline{1}$$

- Entropy:

$$s\eta = -\frac{\partial\psi}{\partial\vartheta} = \underline{A}\cdot\underline{\varepsilon} + c\,(\vartheta-\vartheta_0)$$

Basic principles of material modeling

- Physical consistency: in line with axioms of mechanics

  "thermodynamic consistency": entropy inequality fulfilled
- Equipresence: same functional dependencies for all constitutive equations
- Causality: in thermomechanics, everything depends on $\underset{\sim}{u}$ and v
- Local action: dependence from distant points is neglected $\rightsquigarrow$ dependent variables depend on local values ("simple bodies")
- Objectivity: Independence from rigid body rotations and choice of coordinate system
- Material symmetry: Invariance w.r.t. transformations of material symmetry group

Concept of internal variables

- Dissipation: Irreversible transition of potential energy in different form of energy, e.g. heat

- Constitutive equations for isothermal processes

$$\psi := \psi\left(\underline{\varepsilon}, \alpha_1, \dots \underbrace{\alpha_n, \underline{\alpha}_1, \dots \underline{\alpha}_{\tilde{n}}}\right)$$

internal variables → to phenomenologically describe change of microstructure

$$\rightsquigarrow \underline{\sigma} := \underline{\sigma}\left(\underline{\varepsilon}, \alpha_1, \dots \underline{\alpha}_{\tilde{n}}\right)$$

- Additional constitutive equations → evolution equations

$$\frac{d\alpha_i}{dt} = \dot{\alpha}_i := \dot{\alpha}_i\left(\underline{\varepsilon}, \alpha_i \dots \underline{\alpha}_{\tilde{n}}\right)$$

$$\underline{\dot{\alpha}}_i := \underline{\dot{\alpha}}_i\left(\underline{\varepsilon}, \alpha_i \dots \underline{\alpha}_{\tilde{n}}\right)$$

→ Calculation of $\underline{\alpha}$ / α at current time requires time integration of evolution equations

→ initial values → $\alpha\left(t = t_0\right), \underline{\alpha}\left(t = t_0\right)$